

KELVIN TAWIAH

(318) 245-8515 | ktawiah@gmail.com | [LinkedIn](#) | [Portfolio](#)

EDUCATION

Grambling State University

Grambling, LA

B.S. in Computer Science & Mathematics — GPA: 3.9

Expected Graduation: May 2028

Relevant Coursework: Data Structures, Algorithms, Object-Oriented Programming (Java), Distributed Systems, Operating Systems, Cloud Computing, Computer Networks, Computer Architecture, Machine Learning

TECHNICAL SKILLS

Languages: Java, Python, C++, Go, JavaScript, TypeScript, SQL, Bash

Frameworks & Tools: Spring Boot, React, Node.js, Django, Redis, PostgreSQL, MongoDB, JUnit, Pytest

Cloud/Infra: GitHub Actions, AWS (EC2, Lambda, S3, CloudWatch), GCP, Azure, Docker, Kubernetes, Terraform, Jenkins

Systems & Concepts: System Design, CI/CD, TDD, Agile (Scrum, Standups), Testing (Unit, Integration, Load), Concurrency, Observability, REST/gRPC APIs

EXPERIENCE

Campus Hustle

Apr 2025 – Present

Software Engineer, Backend Infrastructure

- Contributed to a student entrepreneurship platform, implementing **Kafka-based messaging** and **API Gateway** for service routing.
- Integrated **rate limiting** and **circuit breaker** patterns using Resilience4j and documented **REST APIs** with OpenAPI/Swagger.
- Configured **CI/CD** pipelines with **GitHub Actions**, using **CloudWatch** for real-time logging, alerting, and operational monitoring.
- Hosted services using **AWS EC2** with **load-balanced** instances secured via **VPC** network isolation and IAM-based access control.

Google

Feb 2025 – May 2025

Software Engineering Fellow

- Completed a 10-week fellowship with 1:1 mentorship in **algorithms**, **systems design**, and **distributed systems**.
- Applied **iterative refactoring**, **amortized analysis**, and **greedy heuristics** in **Java** to solve weekly optimization challenges.
- Designed an ad click aggregator using **TTL caching**, **RabbitMQ queues**, and **sharded PostgreSQL** cluster for impression events.
- Analyzed interactions between replicated ad data and serialized attribution writes to ensure data integrity under high concurrency.

Bloomberg

Jul 2024 – Sept 2024

Software Engineering Accelerator

- Developed a **portfolio manager** in **Next.js** and **TypeScript**, enabling users to simulate trading with a \$10,000 virtual balance.
- Integrated secure authentication via **Auth.js**, with API-layer validation, session management, and input sanitization.
- Practiced **TDD** and **Agile** development through standups, retros, and weekly pair programming with Bloomberg engineers.

JPMorgan Chase

Feb 2025

Data Analyst, Data For Good Hackathon

- Built a **data processing pipeline** using **pandas**, **NumPy**, and **SQLAlchemy** to aggregate 100K+ Title I school records.
- Created a **logistic regression classifier** using scikit-learn and **k-fold cross-validation** to identify high-risk underfunded schools.
- Leveraged **GeoPandas** and **zip code mapping** to correlate funding gaps with regional factors and visualize district-level disparities.

PROJECTS

QuantaFs — Distributed File System | *Java, gRPC, RocksDB, Kubernetes, Raft Consensus*

- Developed a **multi-node data store**, leveraging the **Raft consensus algorithm** for leader election and log replication across nodes.
- Built **microservices** in Java with gRPC for inter-node communication; used RocksDB for efficient local persistent storage.
- Implemented **multi-threading** and **concurrency controls** to ensure data consistency and high throughput under load.

ChapterCircle — Hybrid Book Recommender Engine | *Spring Boot, NLTK, MongoDB, Google Books API*

- Built a **hybrid recommender engine** that suggests books based on **genre similarity** and **sentiment analysis** of user reviews.
- Integrated **OpenAI's embeddings API** with **Pinecone** vector DB for **semantic search**; fine-tuned **emotion classification** for review rankings based on sentiment intensity.
- Ingested a **10K+** books from Google Books API; used MongoDB **sharding** and **indexing** for metadata queries and fallback filtering.

NeuroSync — Emergency Stroke Triage System | *Django, PostgreSQL, Celery, AWS*

- Developed a stroke triage platform with asynchronous alerts via **Twilio SMS**, using **Celery workers** with Redis-based task queues.
- Leveraged **Redis caching**, **PostgreSQL indexing**, and **Django ORM profiling** for high-concurrency EHR simulations.
- Deployed with **Nginx + Gunicorn** on **AWS EC2** instances behind an Elastic Load Balancer for high-availability under stress.

LEADERSHIP & ACTIVITIES

Vice President, ACM (GSU Chapter)

Jan 2025 – May 2025

- Expanded active membership from **17** → **53** by leading technical workshops, interview prep, and resume review sessions.
- Organized campus-wide hackathons and tech talks, engaging **100+** **participants** and local industry professionals.

Additional Involvements: CodePath, ColorStack, Google Summer of Code (GSoC), NSBE, NVIDIA Summer Bridge, TMCf Scholar